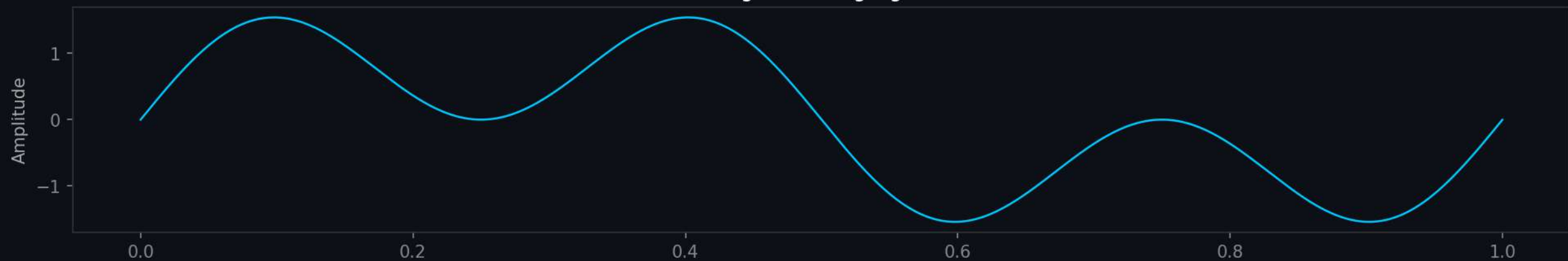
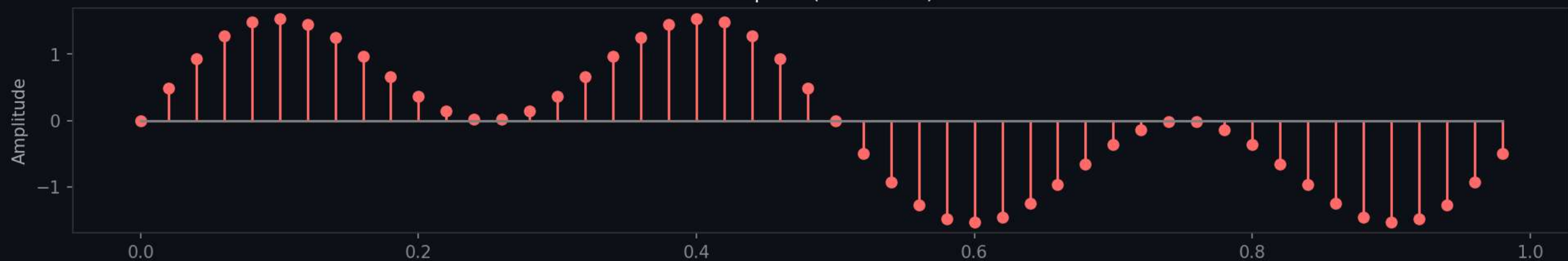


Block A — Sampling & Quantization

Original analog signal



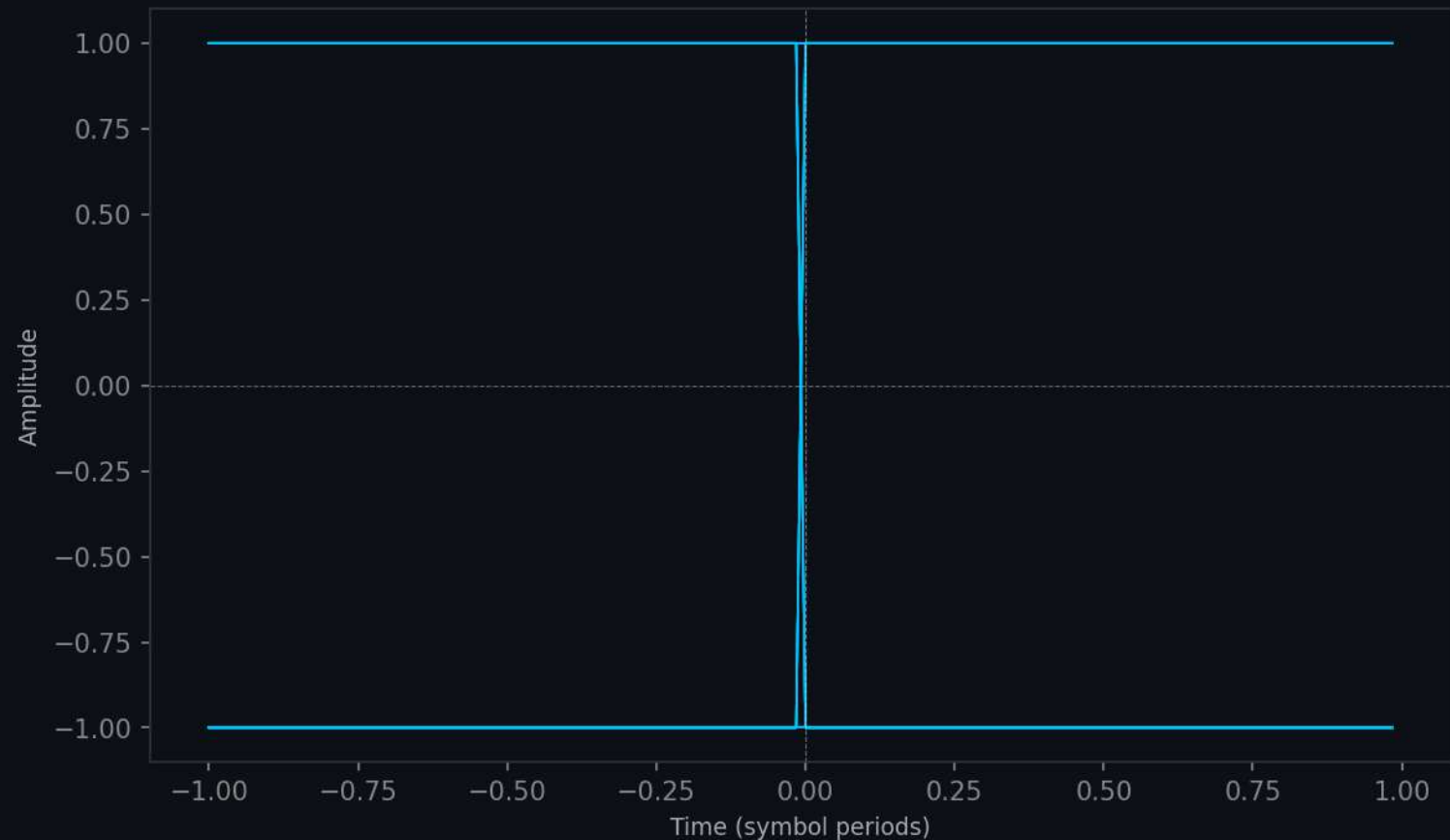
Sampled ($f_s = 50$ Hz)



Quantized ($L = 16$, SQNR = 24.7 dB)



Eye Diagram — POLAR (rectangular NRZ)

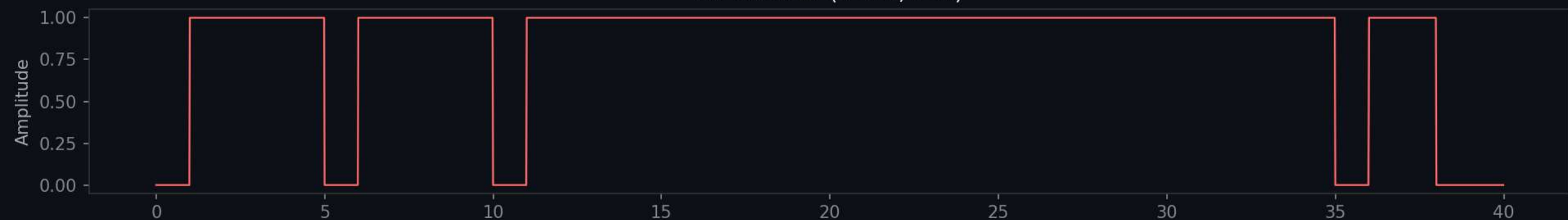


Block B — Line Coding Waveforms (first 40 bits)

Polar NRZ (1→+A, 0→-A)



On-Off OOK (1→+A, 0→0)

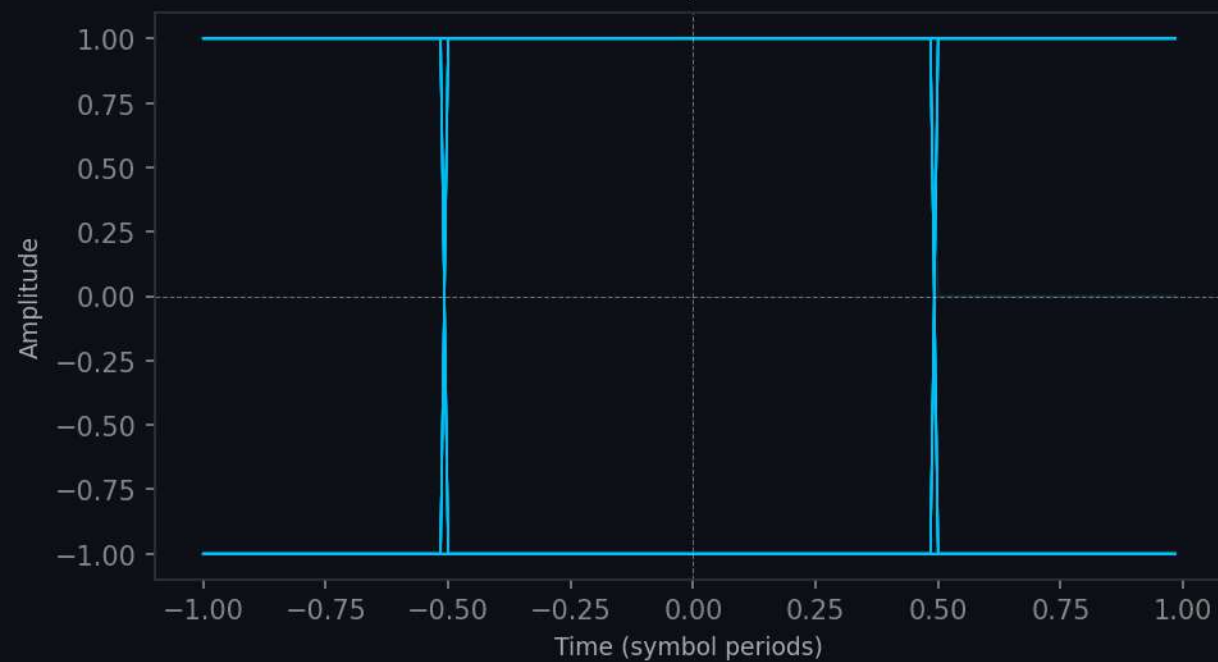


Bipolar AMI (1→±A alternating, 0→0)

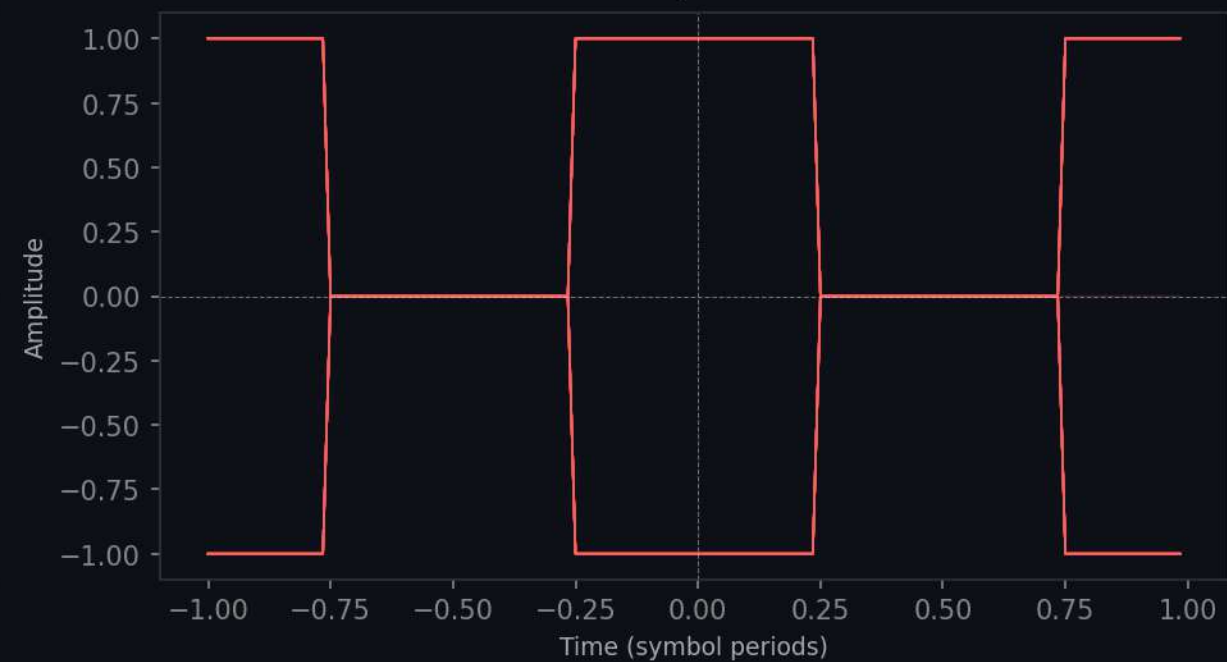


Block C — Eye Diagrams (all pulses)

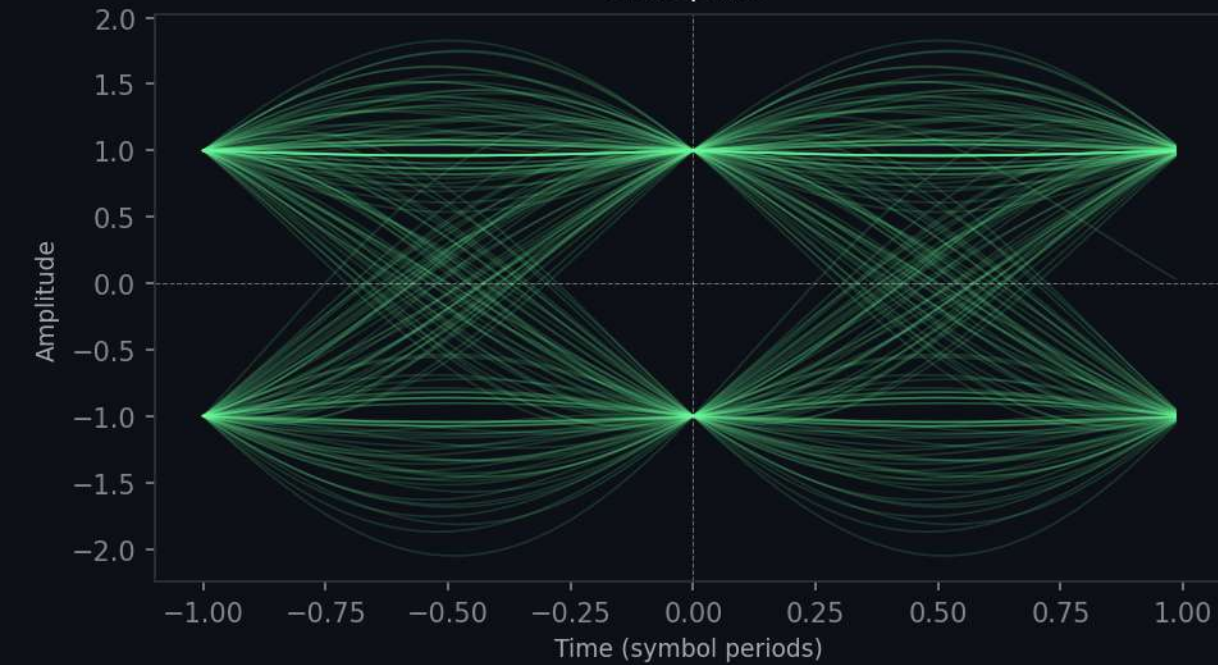
NRZ pulse



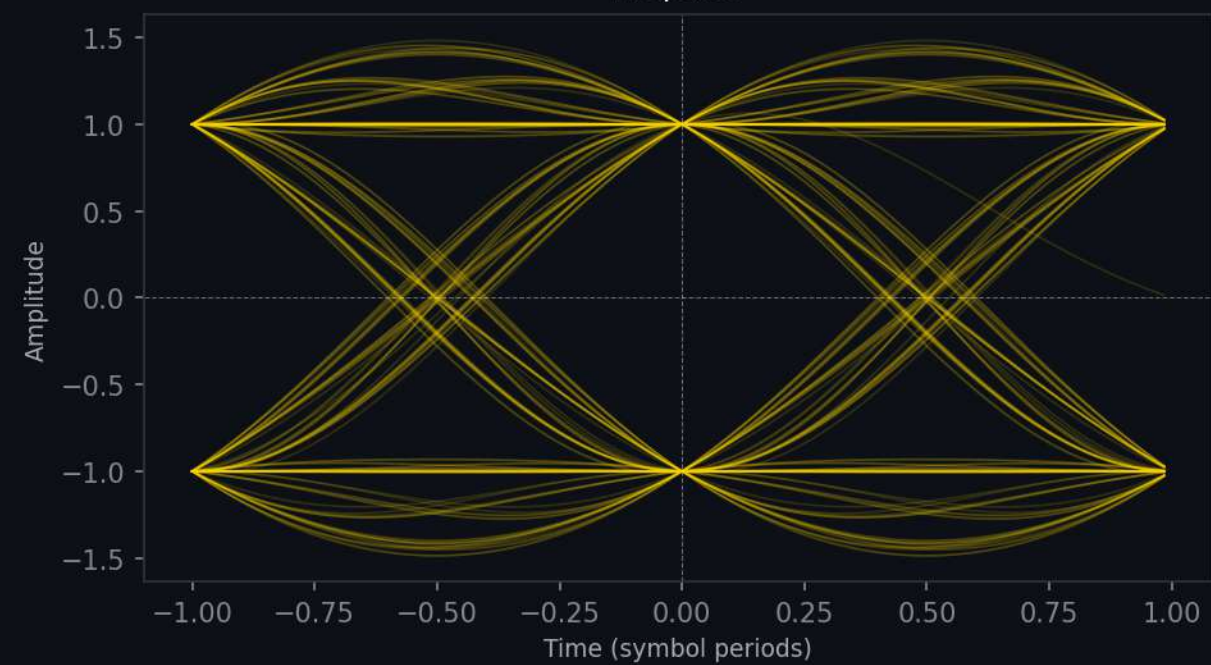
RZ pulse



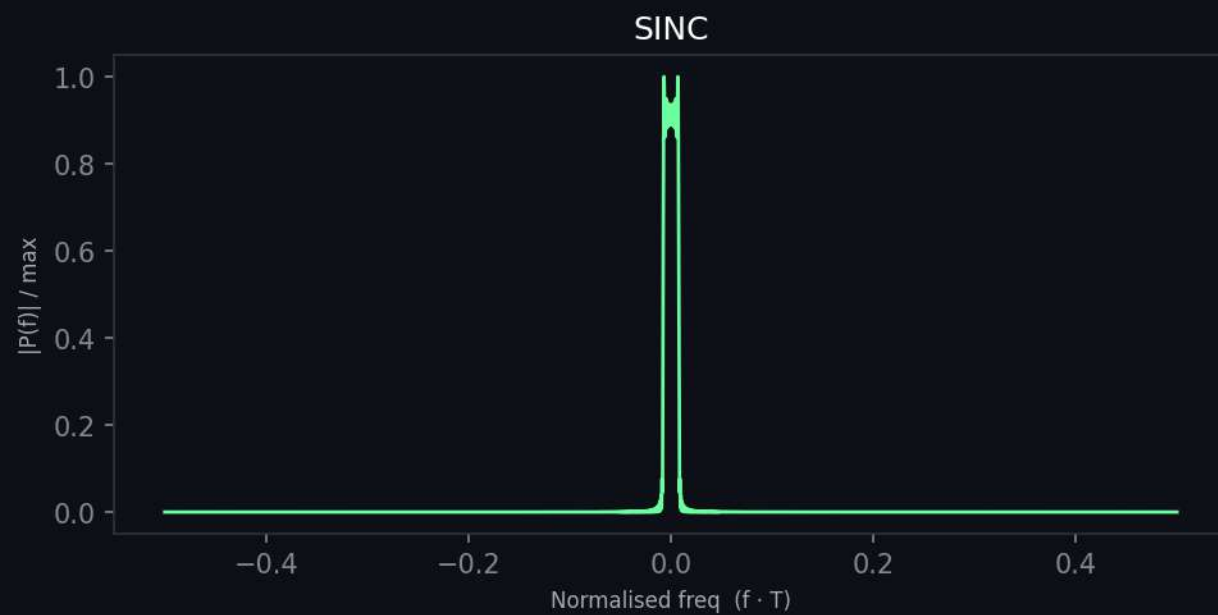
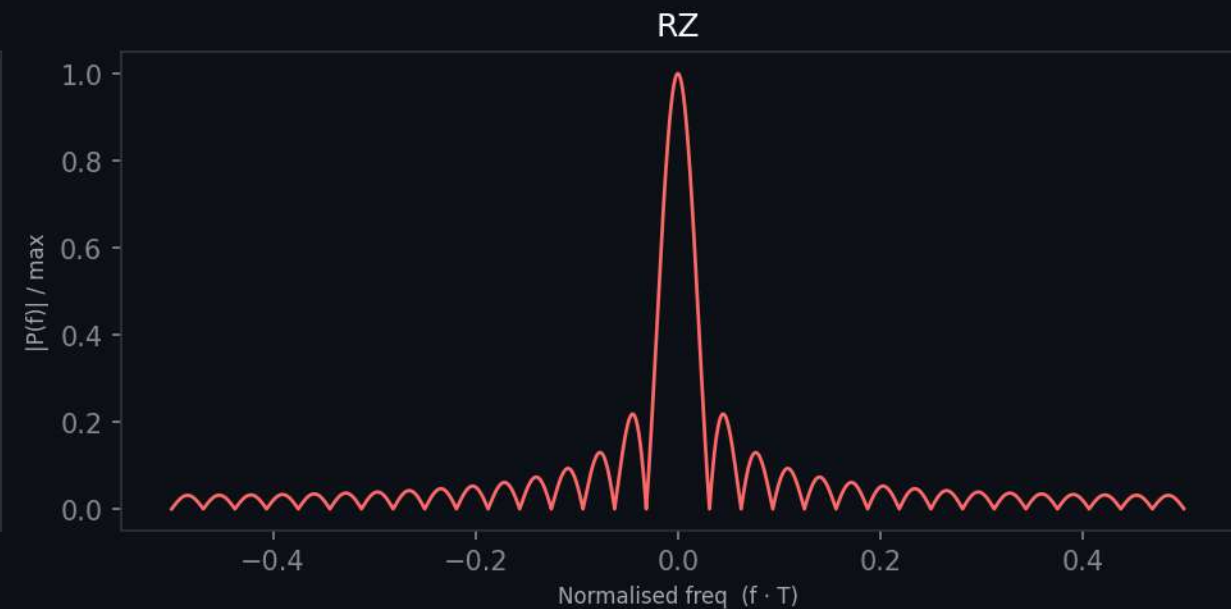
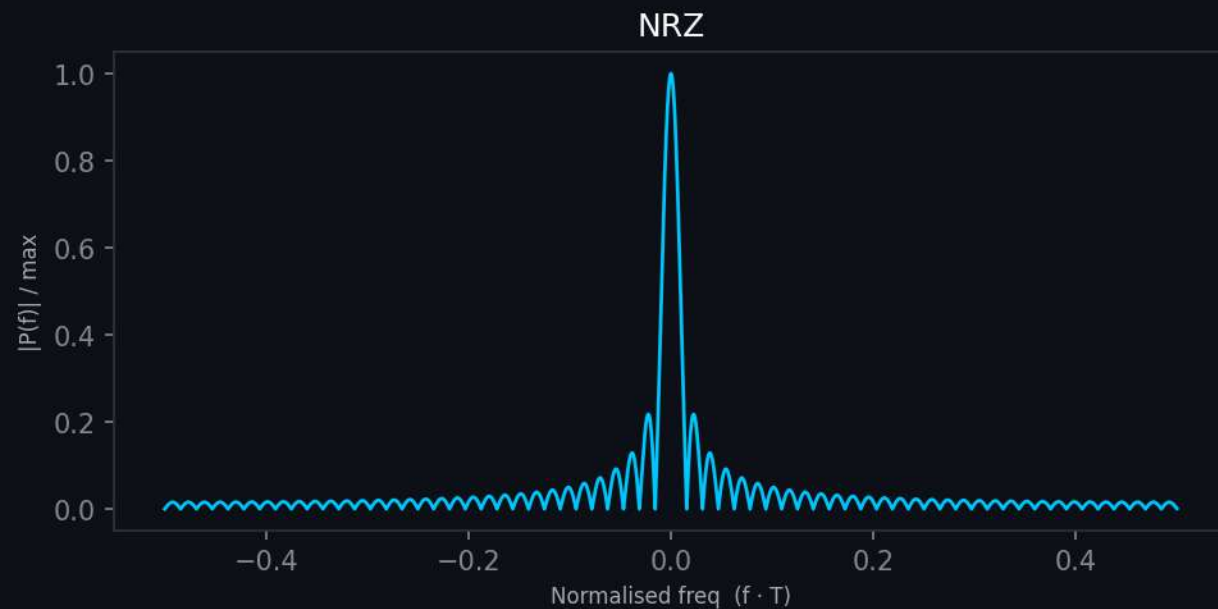
SINC pulse



RC pulse

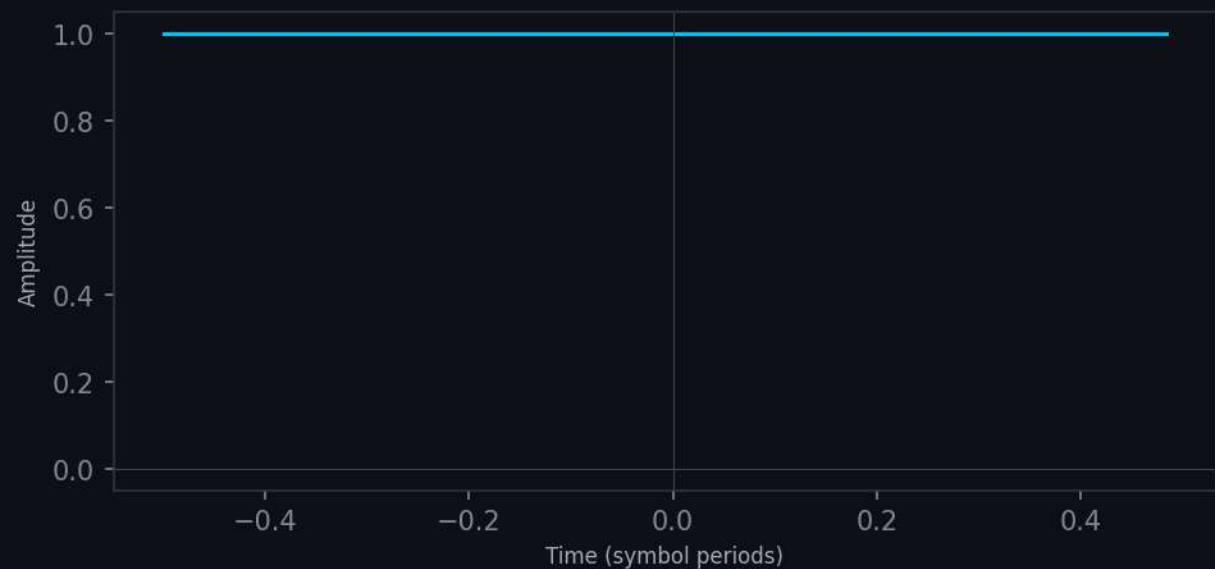


Block C — Pulse Shapes (Frequency Domain)

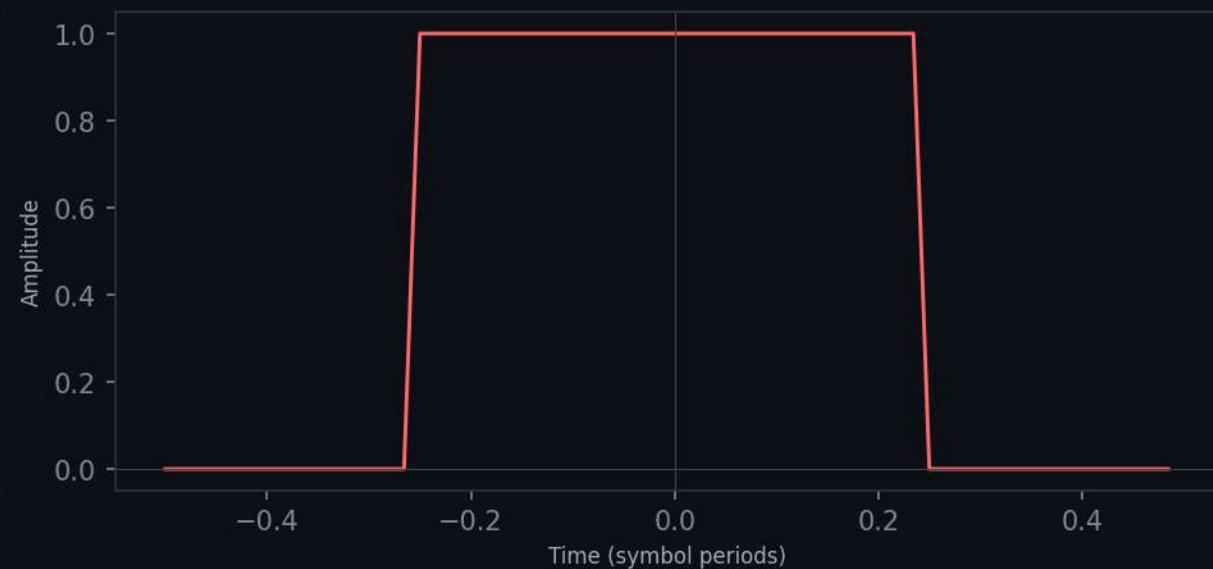


Block C — Pulse Shapes (Time Domain)

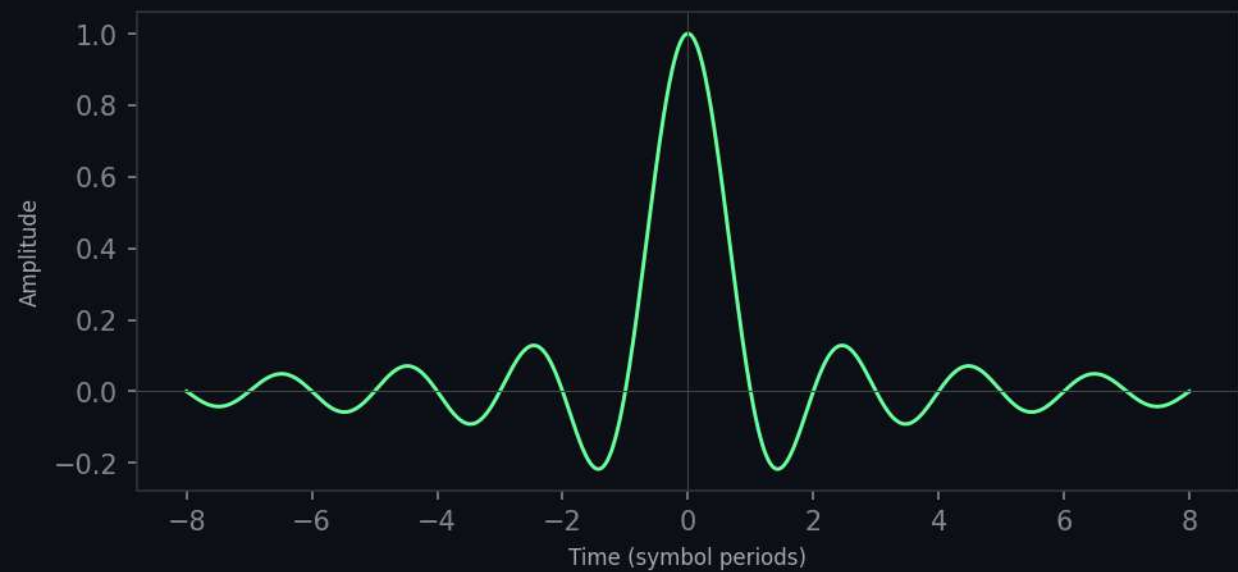
NRZ



RZ



SINC

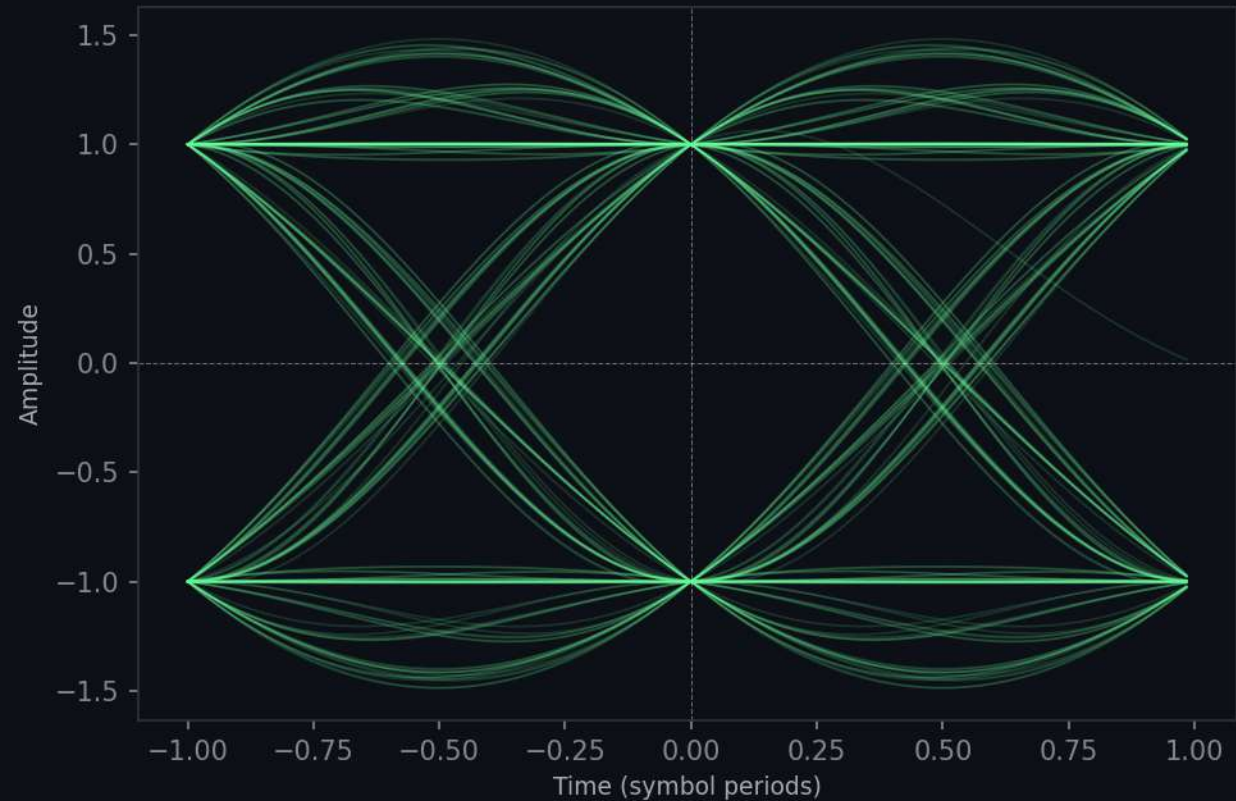


RC ($\beta=0.5$)

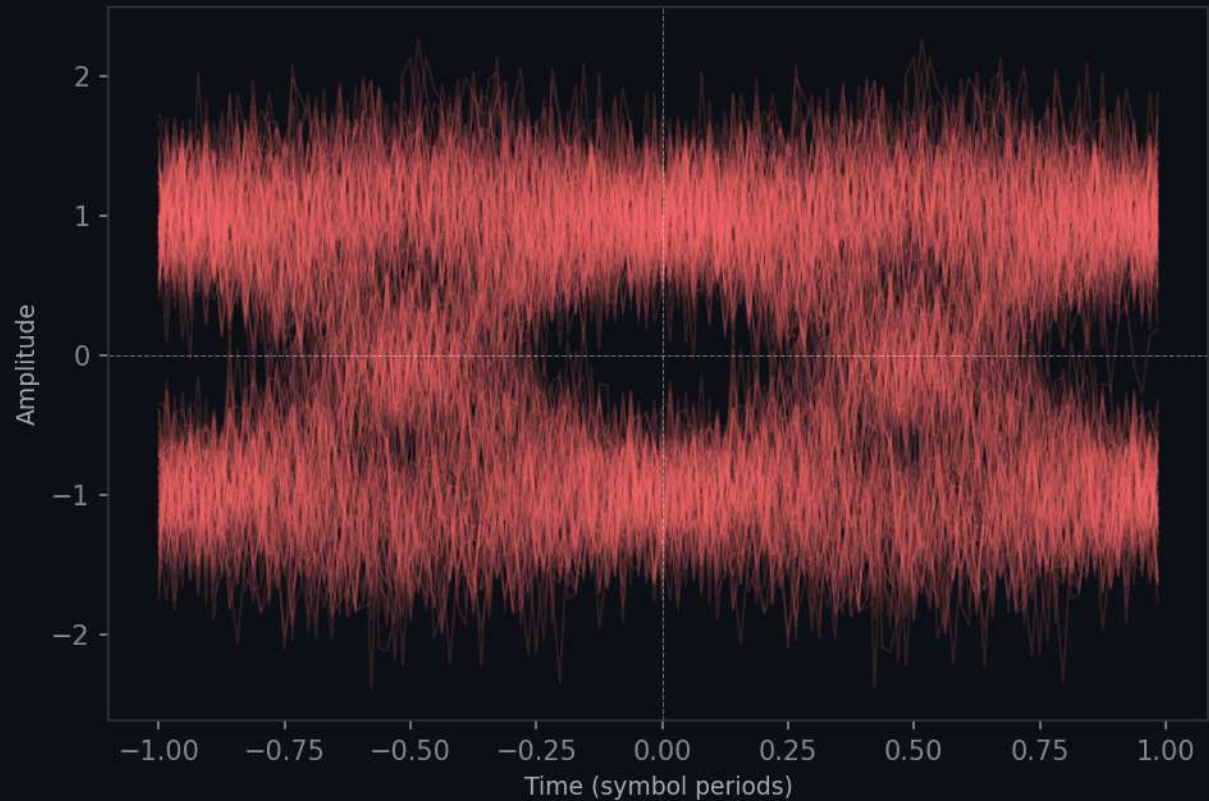


Block D — AWGN Channel (SNR = 10.0 dB)

Eye — Before Channel (RC shaped)

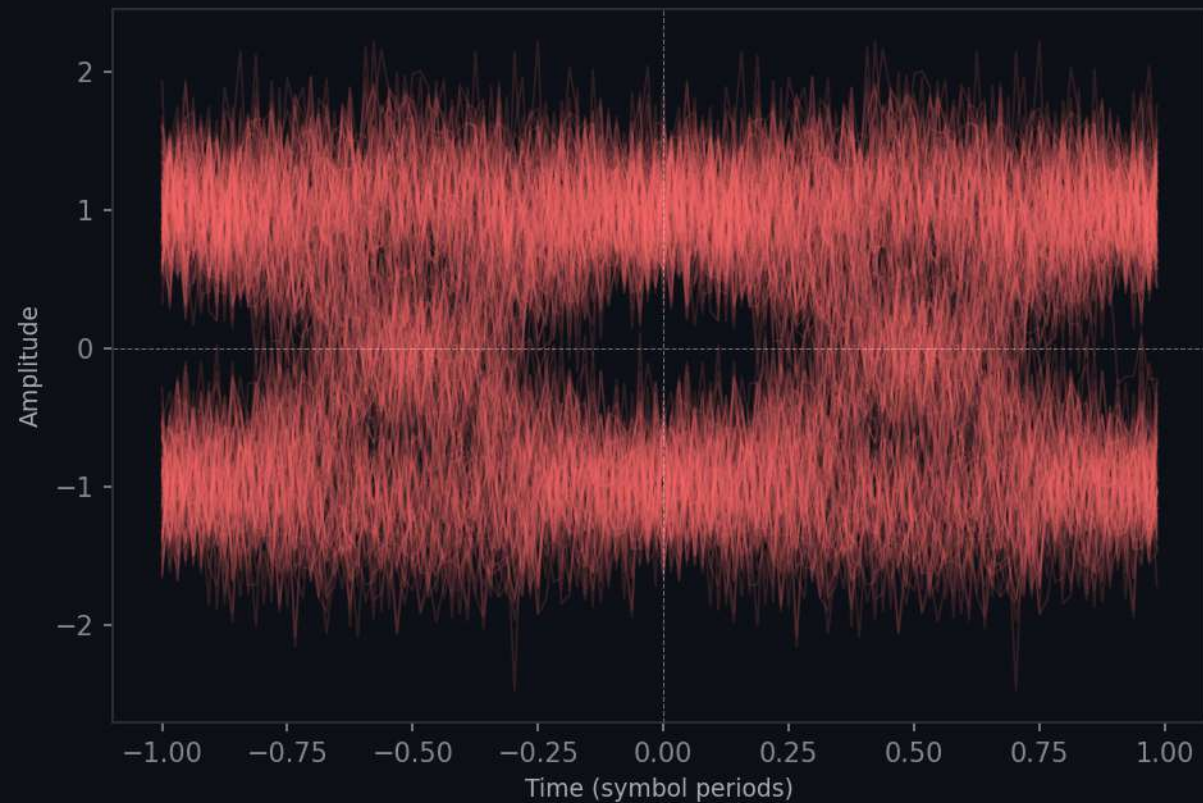


Eye — After AWGN (SNR = 10.0 dB)

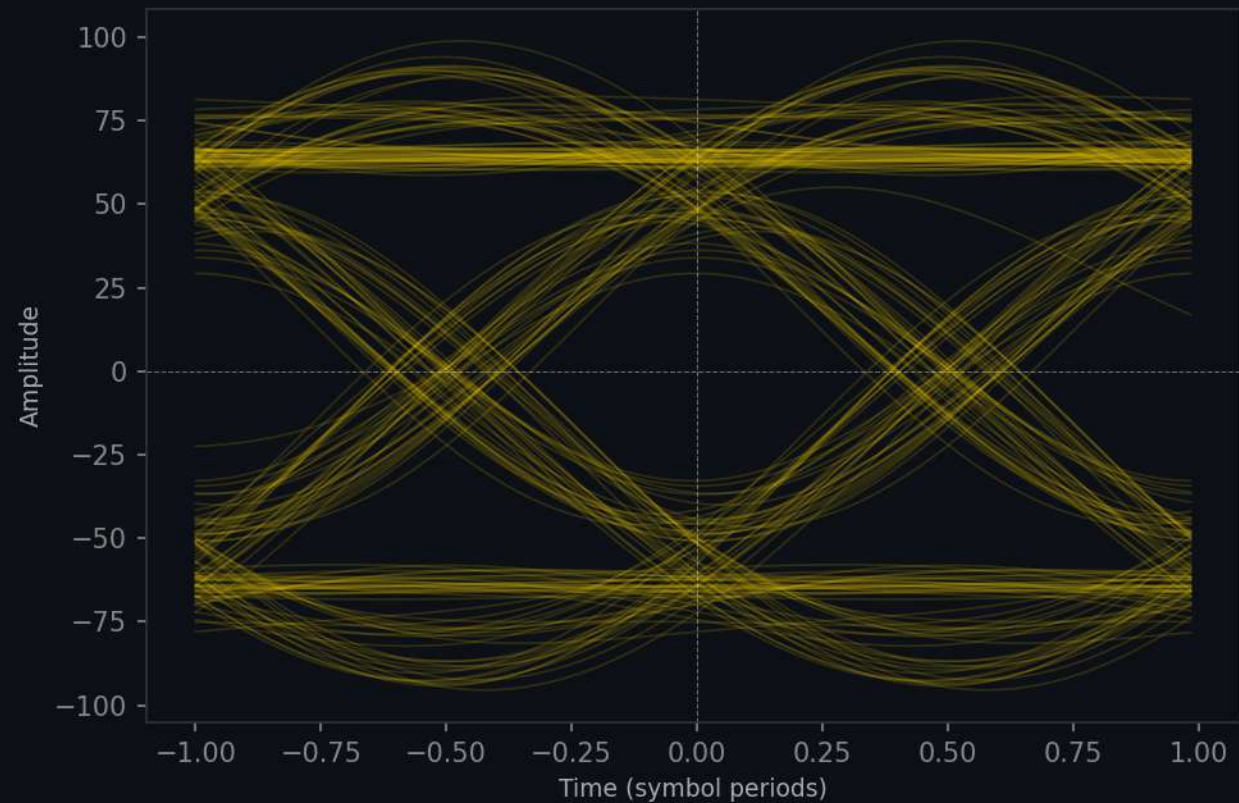


Block E — Matched Filter

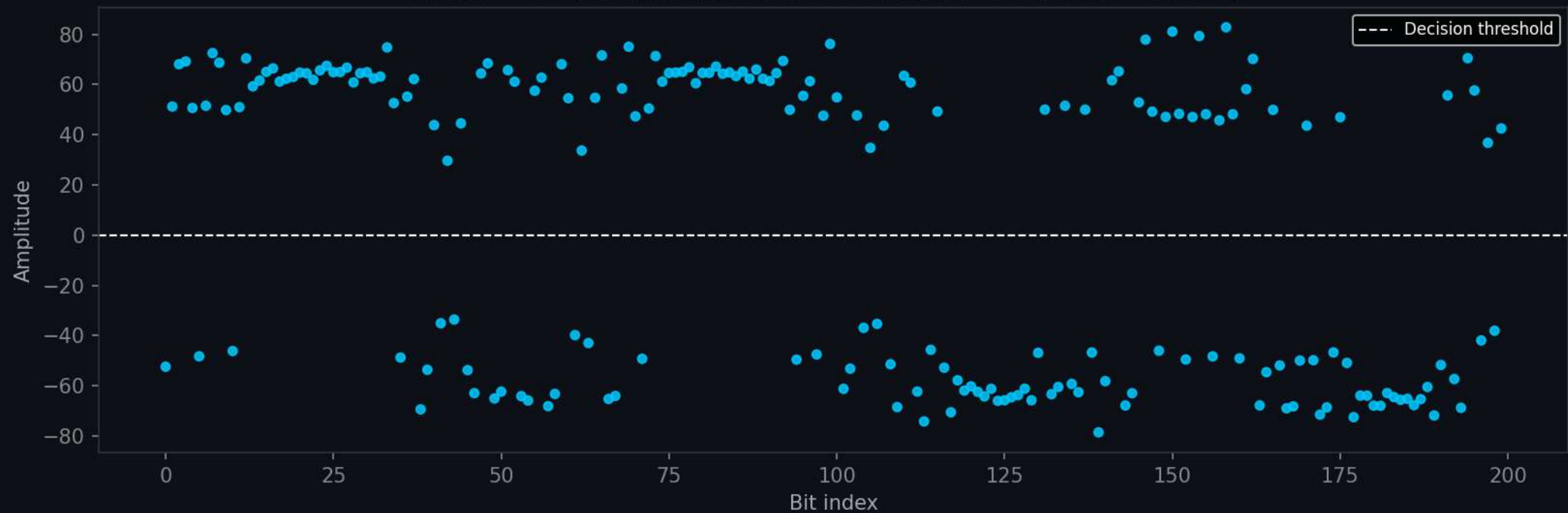
Eye — Receiver Input (noisy)



Eye — After Matched Filter



Block F — MF Output Samples (cyan=correct, red=error) BER = 0.00e+00



Block G — BER vs SNR (Waterfall Curve)

